Institute of Technology of Cambodia

Department Of Industrial and Mechanical Engineering

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Subject: Lathe Machine
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Lathe Machine

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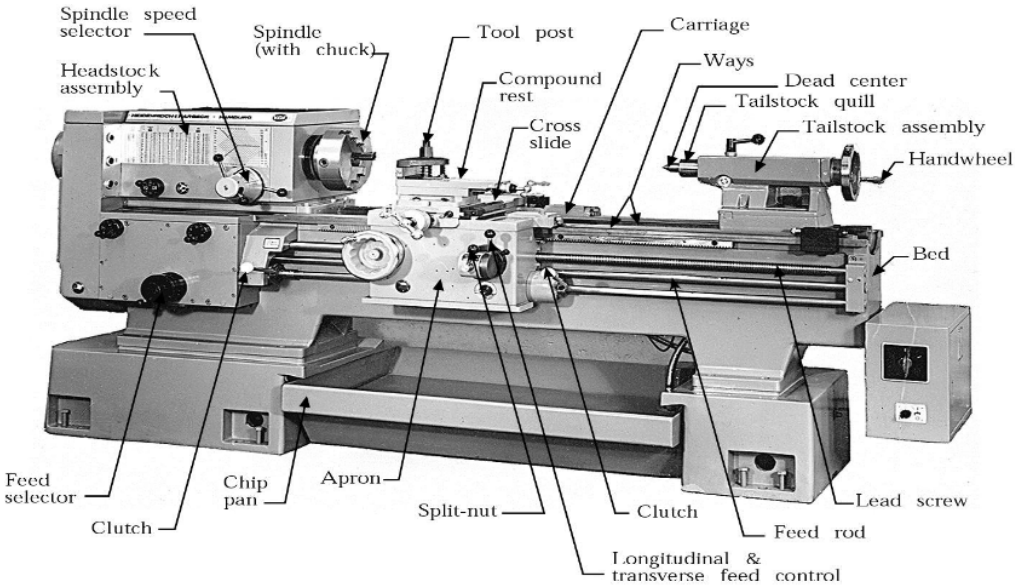
Lathe Machine

I. Introduction

A lathe is a machine that removes metal from a piece of work to the required shape and size. The origin of turning lathes goes back to around 1300 BC when the Egyptians first developed a two-person lathe. One person turned the wooden workpiece with a rope while the other used a sharp tool to cut shapes in the wood. The Romans improved the Egyptian design with the addition of a turning bow. Early bow lathes were also developed and used in Germany, France, and Britain. In the Middle Ages, a pedal replaced hand-operated turning, freeing the craftsman's hands to hold the woodturning tools. The pedal was usually connected to a pole, often a straight-grained sapling. Today's system is called the "spring pole" lathe (Pole-lathe). During the Industrial Revolution, mechanized power was applied to the lathe via steam engines and line shafting, allowing faster and easier work. Between the late 19th and mid-20th centuries, individual electric motors at each lathe replaced line shafting as the power source. Beginning of the 1950s, servomechanisms were applied to the control of lathes and other machine tools via numerical control (NC).

A lathe machine rotates a workpiece while a stationary cutting tool removes material. The machine supports multiple operations, including adjusting **speed, feed, and tool position**. the lathe allows precise machining of cylindrical or conical components.



II. Objective

- Understand the basic operations of a lathe machine
- Learn how to use tools for turning, facing, and drilling
- Observe safety procedures

III. Equipment and Procedure

3.1. KNUTH Werkzeugmaschinen GmbH Lathe, Germany

❖ Equipment

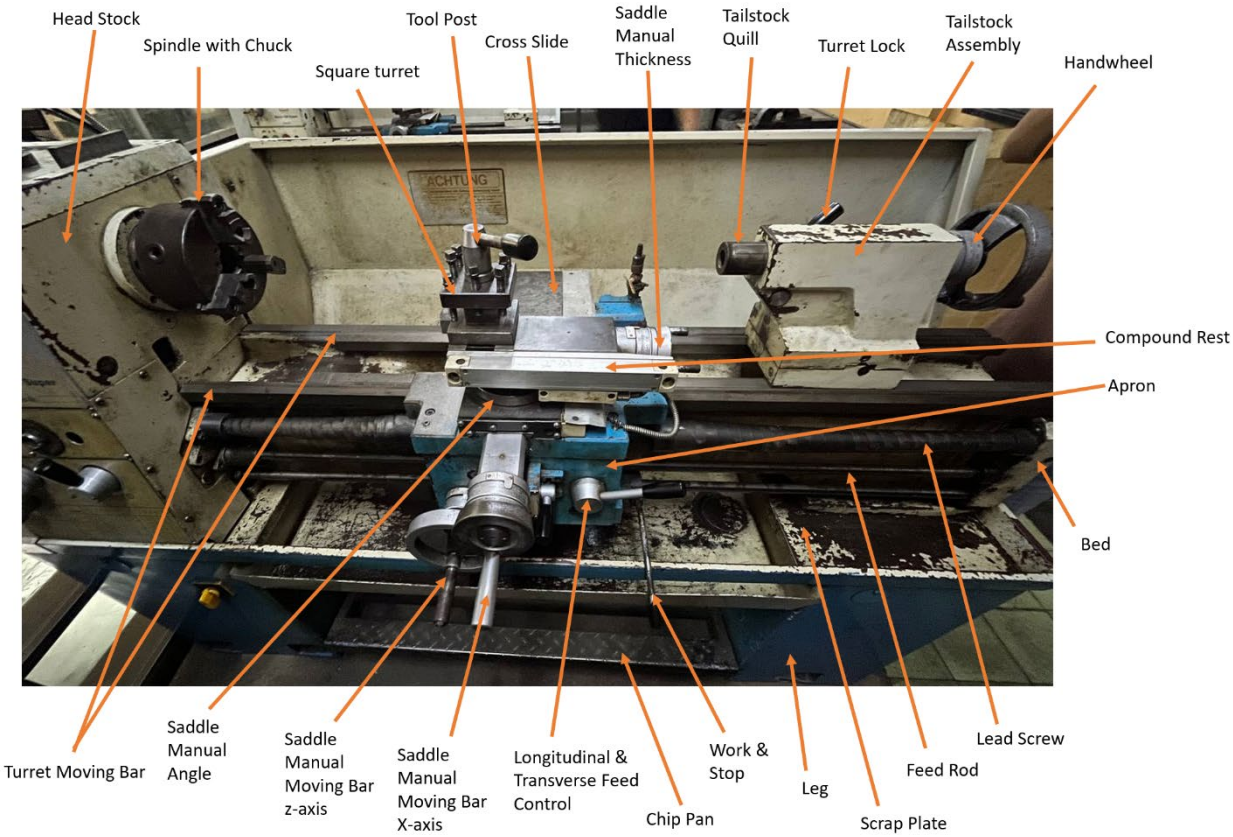
The Lethe machine has many types, depending on the year and company that made it. Now, we will talk about the machine that we use in our workshop, which has belonged **to KNUTH Werkzeugmaschinen GmbH, Germany, since 2002**. The type of this machine is KNUTH Basic 180 Super in 2002.

Here are the published specs for the KNUTH Basic 180 Super:

- **Swing over bed:** 356 mm (14″)
- **Swing over cross-slide:** 220 mm (9″)
- **Distance between centers:** approx. 1,000 mm (39″)
- **Spindle bore:** 38 mm (1.5″)
- **Spindle speed range:** 45–1,800 RPM (mechanical gearbox)
- **Motor power:** 2.4 kW (3.2 HP) rated; 1.5 kW continuous
- **Bed width:** 206 mm (8″), cast-iron, ribbed, induction-hardened guideways
- **Feeds:**
    - Longitudinal (X): 0.015–0.22 mm/rev
    - Longitudinal (Z): 0.043–0.653 mm/rev
- **Threading options:** Metric (0.4–7 mm pitch, 37 options) and Whitworth (4–56 TPI, 28 options)

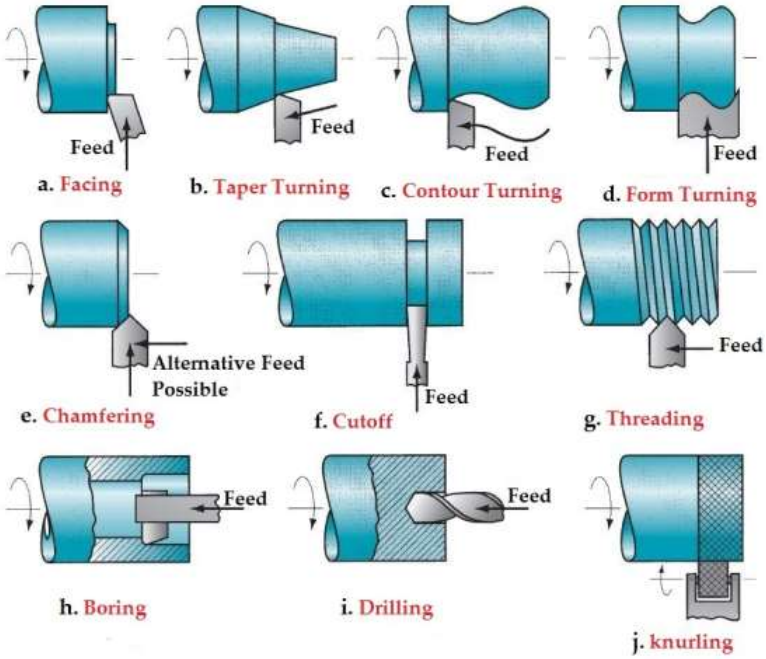
- **Tailstock:** Quill diameter 45 mm, stroke 120 mm, taper MT3, ±10 mm adjustment
- **Weight:** ~880 kg (1,936 lb)
- **Dimensions:** ~1.95 m × 0.75 m × 1.55 m

Image 1: Mechanical Components of the Lathe Machine

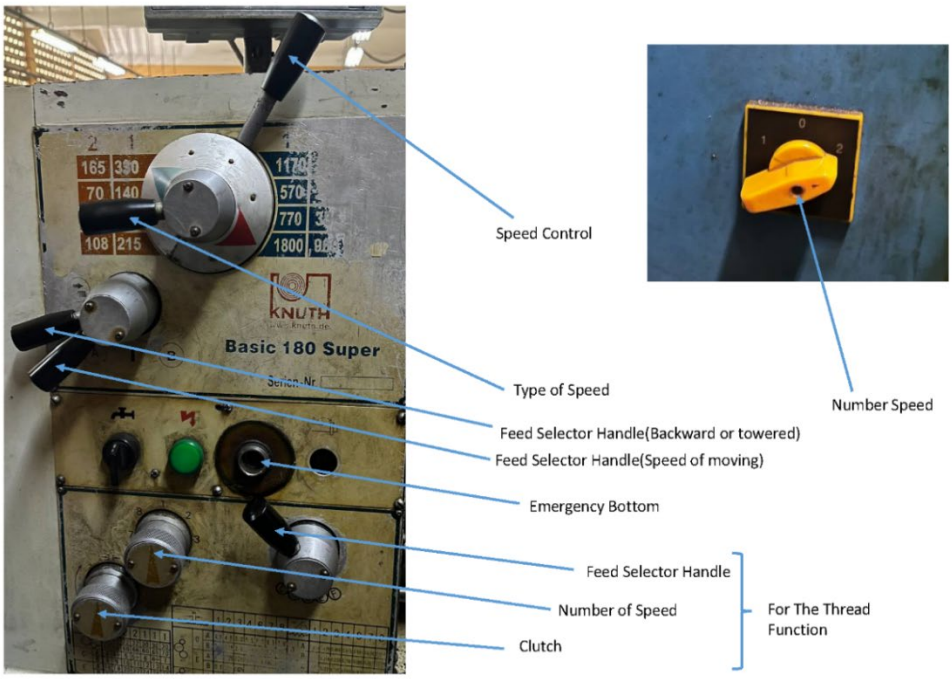


This image labels all the essential mechanical parts involved in the **turning process**. Here's a quick breakdown of key elements:

- **Headstock**: Contains the spindle and gear mechanisms. It holds and rotates the workpiece.
- **Spindle with Chuck**: Holds the workpiece firmly in place.
- **Tool Post / Square Turret**: Holds the cutting tool.
- **Cross Slide & Compound Rest**: Allow tool movement perpendicular and angular to the axis.
- **Saddle & Manual Moving Bars (X, Z-Axis)**: Controls tool carriage along bed.
- **Tailstock Assembly**: Used for supporting long workpieces or holding tools like drills.
- **Bed**: Provides a foundation for the entire lathe.
- **Lead Screw / Feed Rod**: Powers the carriage and cross-slide movement for threading and feeding.
- **Chip Pan & Scrap Plate**: Collect metal chips during operation.
- **Handwheel, Control Levers**: Allow manual adjustment.
- **Lathe Operation**: used for a specific purpose in shaping a workpiece.
- **Cutting Oil**: used for cooling, Lubrication, improved surface Finish, Chip Removal, and Prevention of Rust.



❖ Image 2: Control Panel and Speed Settings



This part focuses on the **lathe's control interface**, specifically for **speed and feed regulation**:

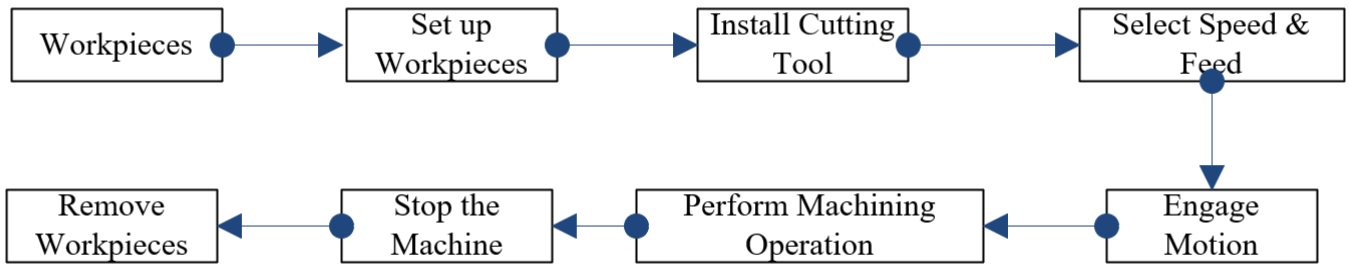
- **Speed Control**: Adjusts spindle rotation speed.
- **Type of Speed**: Switches between different ranges (e.g., low/high gears).
- **Feed Selector Handles**:
  - One changes the feed direction (forward/reverse).
  - Another adjusts the feed rate (speed of movement).
- **Emergency Button**: Instantly stops machine operation.
- **Number Speed Selector**: Chooses specific RPM (revolutions per minute).
- **Clutch**: Engages or disengages power transmission.
- **Thread Function Section**: Controls feed settings specific for threading operations.

❖ Procedure

The material that we can take to be workpieces is:

Material Type	Examples	Machinability	Notes
Ferrous Metals	Mild steel, Cast iron	Moderate–Good	Common in mechanical parts
Non-Ferrous Metals	Aluminum, Brass, Copper	Excellent	Smooth finish, easy to cut
Plastics	Nylon, ABS, Acrylic	Very Good	Watch for melting
Wood	Teak, Pine, Maple	Very Good	Use wood lathe tools

This is the procedure that we will discuss in these points:



The workpieces that we use to work this test for make you an example is aluminum and carbon steel, that has 130mm long and dimension of 30mm.

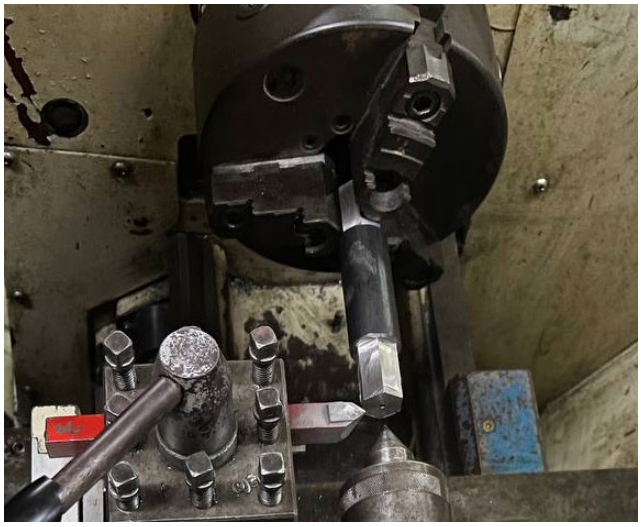
KNUTH Werkzeugmaschinen GmbH Lathe, Germany

1. Set up Workpieces

- Mount the **workpieces** securely in the **chuck** (on the spindle).
- If it's a long piece, you have to drill it and then use **the Tailstock** to support it
- Make sure the specimen is stuck with the chuck and no or minimal vibration when it rotates.

2. Install Cutting Tool

- Choose the right Cutting Tool depending on the material of the workpieces.
- Insert the desired cutting tool into **the Tool post (Square turret)**.
- Adjust the position using the **compound rest** and **cross slide**.
- Make sure **the cutting Tool** stays at the zero point of **the Tailstock.**
- Set the angle of **the Square Turret** depending on your proposal with the work pieces.



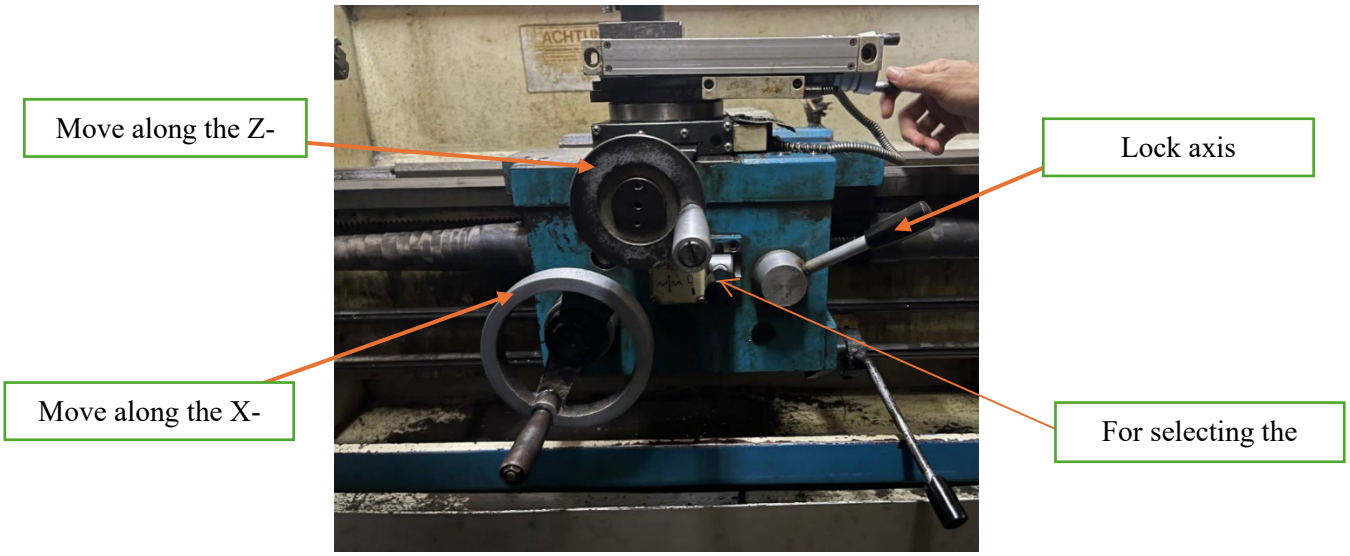
3. Select Speed & Feed

Use the **control panel**:

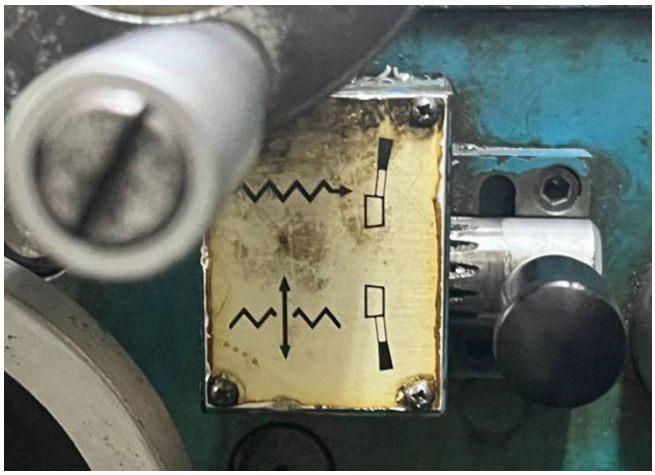
- Choose spindle **RPM** using the **Speed Control** and **Number Speed** selector 1 or 2.
- Use **Feed Selector Handles** to pick feed direction and rate blue or orange.
- Select the right rate of **Lathe Operations** that fits our work pieces

4. Engage Motion

- Use the **Clutch** to start the spindle rotation.
- Move the tool using:
  - **Manual handwheels** (for precision and setup).
  - **Automatic feed rod/lead screw** (for threading and continuous cuts).
  - Move it to the left and turn it down to lock the Z-axis
  - Move it to the right and turn it up to lock the X-axis



For selecting the axis



5. Perform Machining Operation

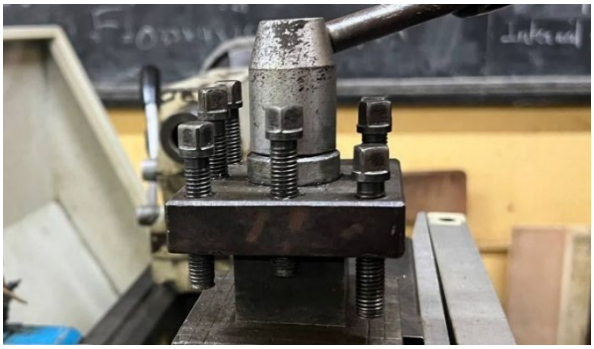
- Control depth with the **cross-slide** and **compound rest (scale is 0.02mm)**.
- Move the tool along the length of the workpiece with the **saddle** or **carriage**.
- Don't forget to open **the Cutting oil or water** to reduce heat and friction of workpieces.

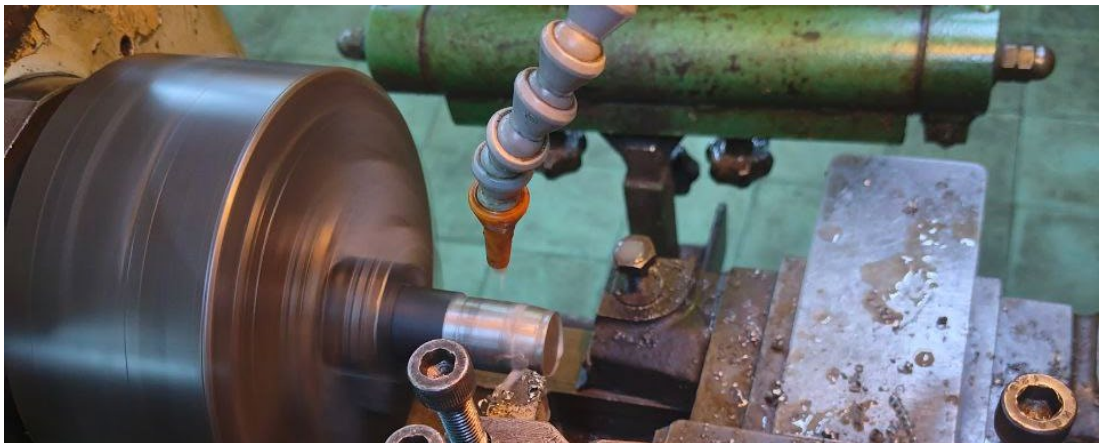
For set the depth that we need to cut

You can do operations like:

- **Facing** (flat cut at the end)

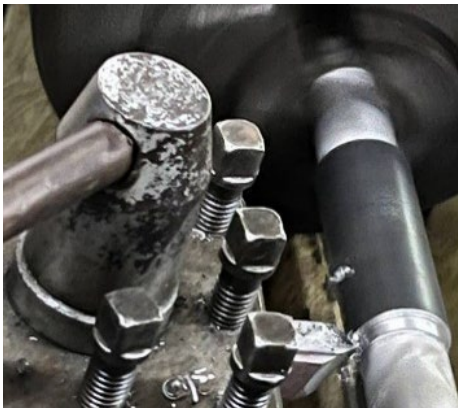
- Set **the Tool Post on the square Turret,** set the angle to for facing
- To make the face smooth and reduce the length of workpieces





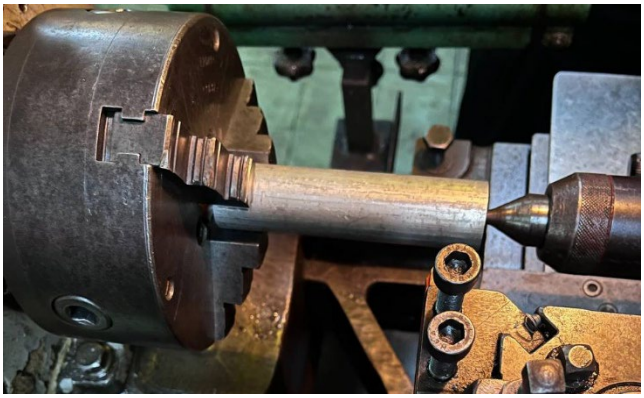
- **Turning** (reducing diameter)

- Use the tailstock to support the workpieces when we want to do turning.
- Lock the position of **the Tailstock**



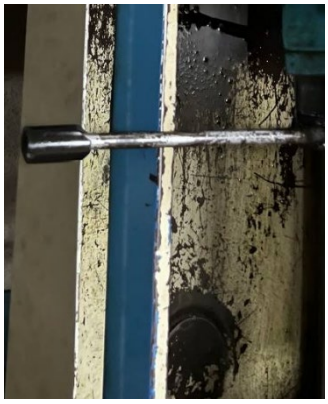
- **Drilling** (with drill bit in tailstock)

- Change the type of the tailstock to the drill bit for drilling.
- Then move it slowly after it touches the workpieces.



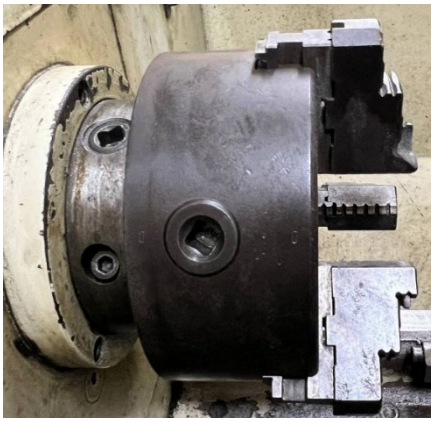
6. Stop the Machine

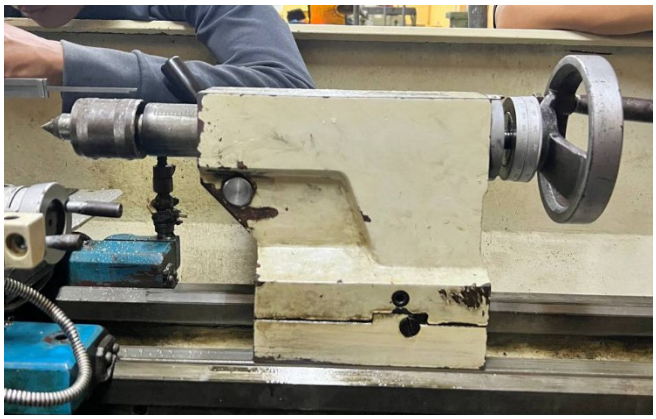
- Move the cutting tool out and far from the workpieces
- Turn **the saddle** up
- Push **the break** down when we want to stop it in the sort of times or emergency times



7. Remove Workpieces

- Remove the workpiece after it fully stops.
- Move **the Tail Stock assembly** out
- Take the specimens out of **the Chuck**
- Use water or oil to clean the specimens





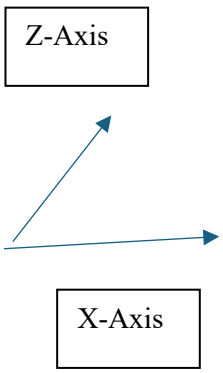
3.2. H. Ernault Somua Lathe, France

❖ Equipment

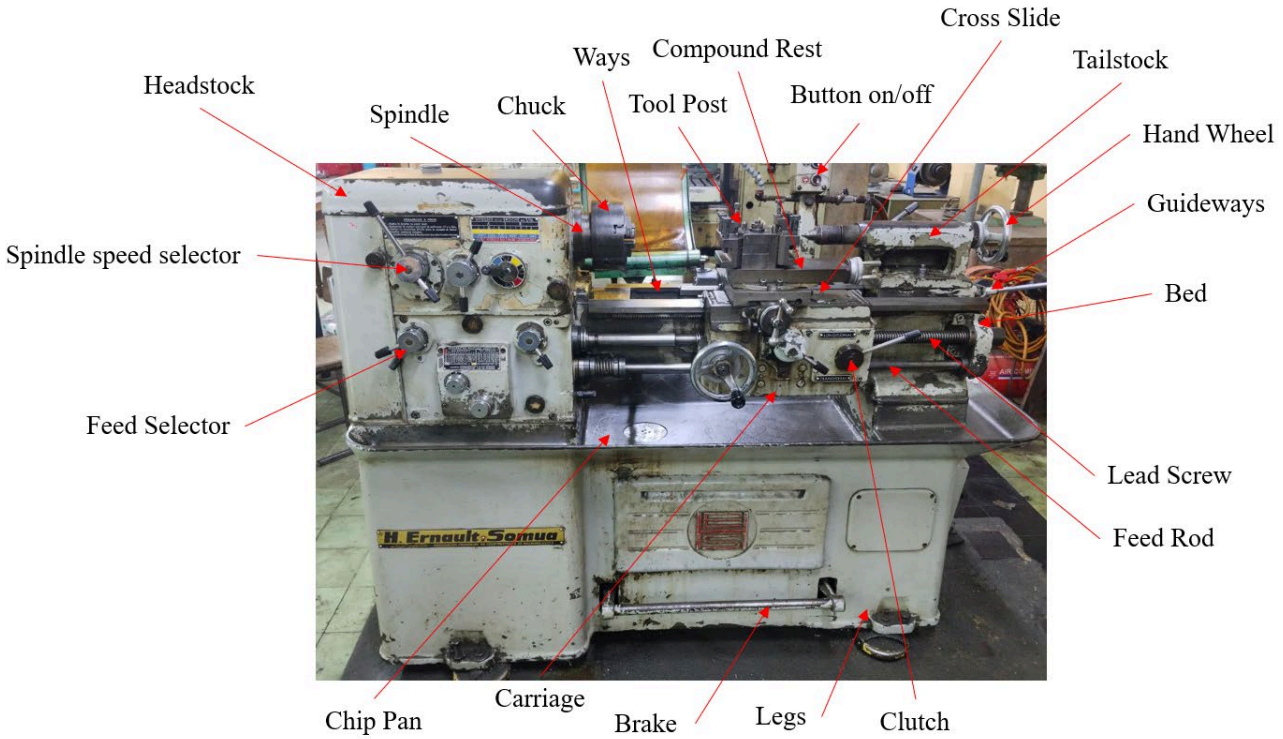
The Lethe machine has many types, depending on the year and company that made it. Now, we will talk about the machine that we use in our workshop, which belongs to **H. Ernault Somua in France, since 1966.** The type of machine is **H. Ernault Somua AC280**:

While exact specs vary depending on production era and region, below is a **typical profile** for Somua lathes of this type:

- **Swing over bed:** ~280 mm
- **Distance between centers:** ~1,000 mm
- **Spindle bore:** ~42 mm
- **Spindle speed:** controlled via belt and gearbox, with multiple step ranges (roughly 50–1,200 RPM)
- **Motor power:** ~3–5 HP (2.2–3.7 kW)
- **Feed rates & threading:** controlled via gear/pulley levers; supports both metric and imperial threads
- **Tailstock:** ~120 mm quill travel, Morse taper (likely MT4–MT5)
- **Feature notes:** Rugged construction, geared headstock, manual feed Bart, trademark Somua pulley levers for speed and thread selection



❖ **Image 1: Mechanical Components of the Lathe Machine**

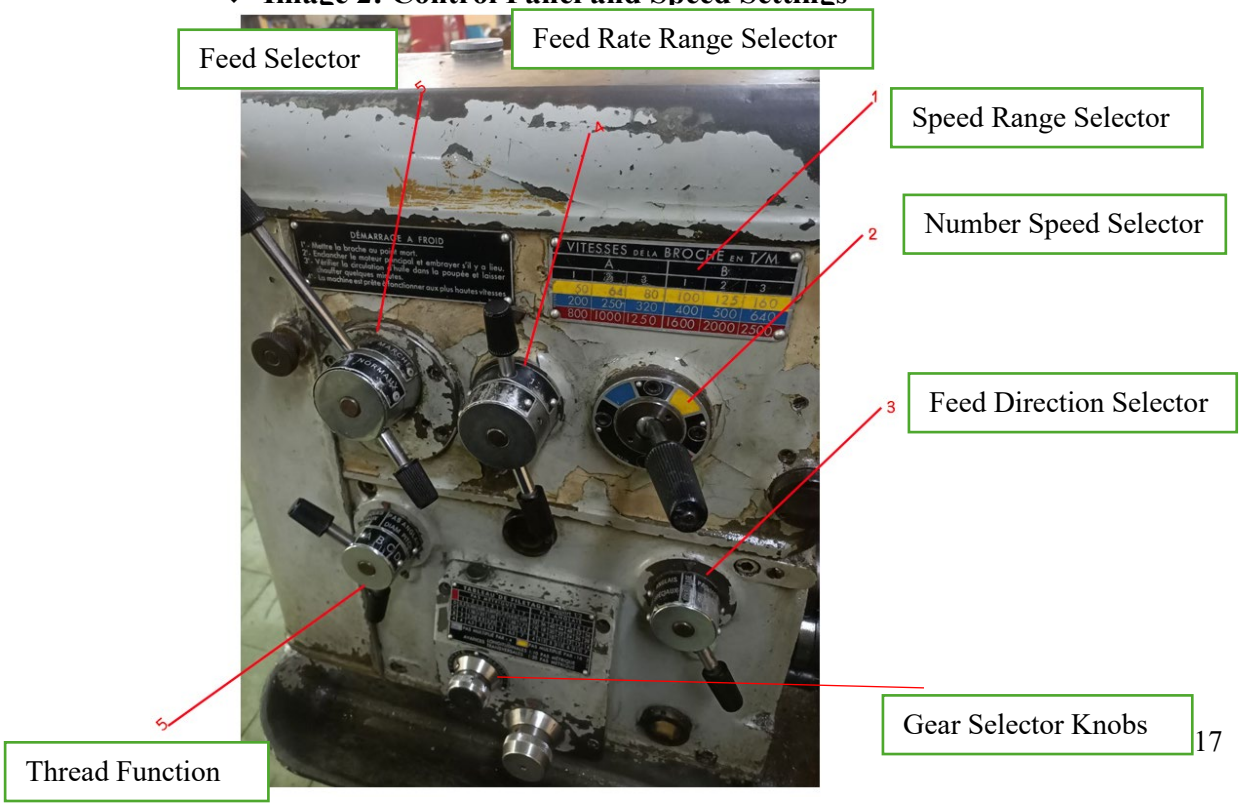


This image labels all the essential mechanical parts involved in the **turning process**. Here's a quick breakdown of key elements:

- **Headstock:** Houses the spindle and the gear system to drive the spindle at various speeds.
- **Spindle:** Rotates the workpiece, driven by the motor through belts or gears.
- **Chuck:** Clamps and holds the workpiece firmly during machining.
- **Spindle Speed Selector:** Adjusts the rotational speed of the spindle by selecting different gear combinations.
- **Feed Selector:** Controls the direction and rate of feed (either longitudinal or cross).
- **Carriage:** Moves the cutting tool along the bed. It supports, moves, and controls the tool post and cross slide.
- **Tool Post:** Holds the cutting tool in position; can be rotated for angular cuts.
- **Compound Rest:** Mounted on the cross slide and used for fine-tuning the tool position or cutting tapers.

- **Ways:** Flat or V-shaped tracks that guide the carriage and tailstock.
- **Cross Slide:** Allows the tool to move perpendicular to the workpiece axis for facing or turning.
- **Tailstock:** Provides support to the free end of the workpiece; can also hold tools like drill bits.
- **Hand Wheel (on tailstock):** Manually moves the tailstock spindle in or out.
- **Guideways:** Precision-ground paths for the movement of carriage and tailstock.
- **Bed:** Base of the lathe that supports all major components and ensures alignment.
- **Lead Screw:** Used for precise movement of the carriage during threading operations.
- **Feed Rod:** Transmits motion from the gearbox to the carriage for automatic feed.
- **Brake:** Stops the spindle quickly for safety or tool changes.
- **Clutch:** Engages or disengages the spindle from the motor.
- **Legs:** Support the entire lathe structure and house the electrical and lubrication systems.
- **Chip Pan:** Collects chips and coolant below the lathe during operation.
- **Button (on/off):** Starts and stops the machine.

❖ **Image 2: Control Panel and Speed Settings**



Here's a breakdown of each labeled component on the lathe control panel and speed settings from the image:

1. **Speed Range Selector**

- Selects the range of spindle RPMs (e.g., low, medium, high).
- Works in combination with the number speed selector to achieve the desired spindle speed.

2. **Number Speed Selector**

- Precisely selects the actual spindle speed in revolutions per minute (RPM), within the selected range.
- You match this to the chart shown on the panel for the correct cutting speed.

3. **Feed Direction Selector**

- Determines the direction of feed: forward or reverse.
- Used for changing the direction of tool movement along the workpiece.

4. **Feed Rate Range Selector**

- Adjusts the range of feed rates—how fast the tool moves relative to the spindle rotation.
- Useful for roughing or finishing operations.

5. **Thread Function**

- Engages the threading mode.
- Enables cutting various thread types and pitches when set correctly.

6. **Feed Selector**

- Selects the type of feed: longitudinal (along the bed) or cross feed (across the bed).
- Determines the path of tool movement.

7. **Gear Selector Knobs**

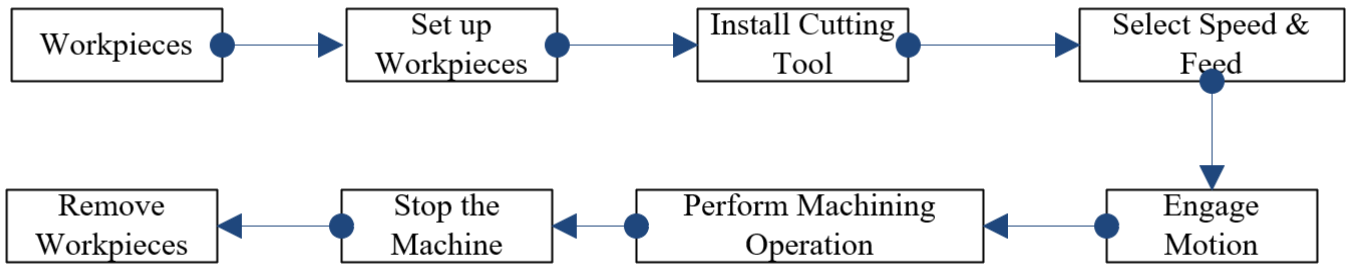
- Change the internal gear combinations to provide different spindle speeds, feed rates, or threading options.
- Work together with the thread and feed controls.

❖ Procedure

The material that we can take to be workpieces is:

Material Type	Examples	Machinability	Notes
Ferrous Metals	Mild steel, Cast iron	Moderate–Good	Common in mechanical parts
Non-Ferrous Metals	Aluminum, Brass, Copper	Excellent	Smooth finish, easy to cut
Plastics	Nylon, ABS, Acrylic	Very Good	Watch for melting
Wood	Teak, Pine, Maple	Very Good	Use wood lathe tools

This is the procedure that we will discuss in these points:

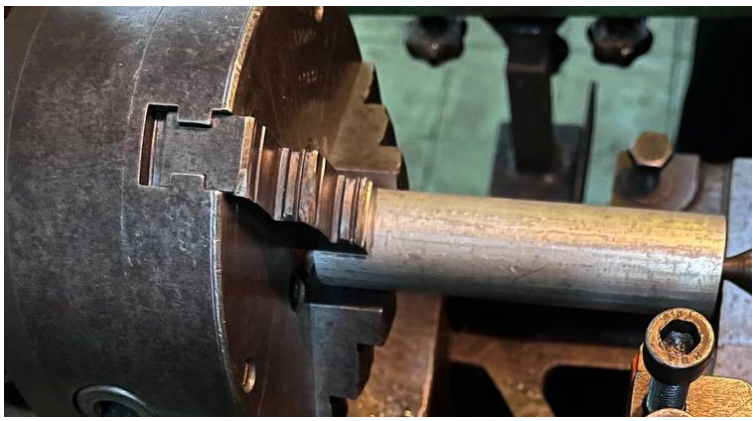


The workpieces that we use to work this test for make you an example is aluminum and carbon steel, that has 130mm long and dimension of 30mm.

H. Ernault Somua Lathe, France

1. Set up Workpieces

- Mount the **workpieces** securely in the **chuck** (on the spindle).
- If it's a long piece, you have to drill it and then use **the Tailstock** to support it
- Make sure the specimen is stuck with the chuck and no or minimal vibration when it rotates.



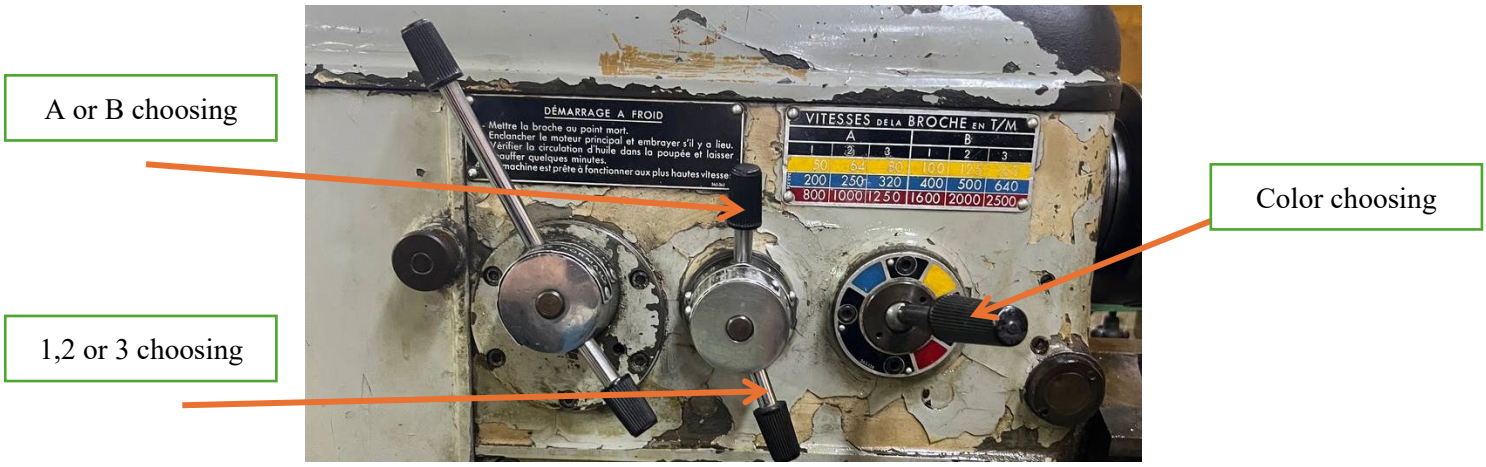
2. Install Cutting Tool

- Choose the right Cutting Tool depending on the material of the workpieces.
- Insert the desired cutting tool into **the Tool post (Square turret)**.
- Adjust the position using the **compound rest** and **cross slide**.
- Make sure **the cutting Tool** stays at the zero point of **the Tailstock.**
- Set the angle of **the Square Turret** depending on your proposal with the work pieces.

3. Select Speed & Feed

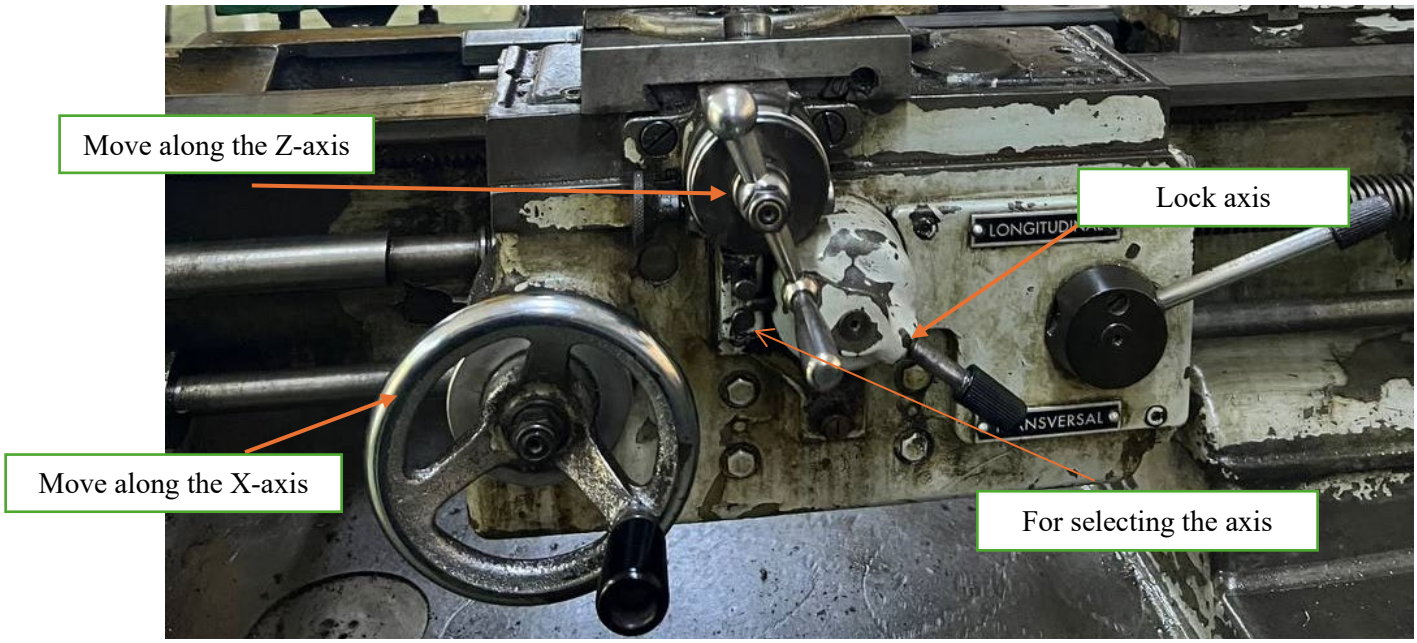
Use the **control panel**:

- Choose spindle **RPM** using the **Speed Control** and **Number Speed** selector A or B in the table of speed rate.
- Select the rate of **Lathe Operations** that fits our work pieces in row number 1, 2, or 3.
- Use **Feed Selector Handles** to pick feed direction and rate blue, yellow, or red.



4. Engage Motion

- Use the **Clutch** to start the spindle rotation.
- Move the tool using:
  - **Manual handwheels** (for precision and setup).
  - **Automatic feed rod/lead screw** (for threading and continuous cuts).
  - The upper is locked on the Z-axis, and then turn **the Lock axis** down
  - The lower is locked on the X-axis, and then turn **the Lock axis** on s



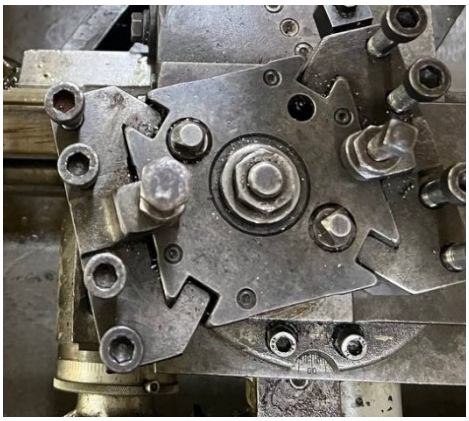
5. Perform Machining Operation

- Control depth with the **(Scale is 0.01mm)**.
- Move the tool along the length of the workpiece with the **saddle** or **carriage**.
- Don't forget to open **the Cutting oil or water** to reduce heat **cross-slide** and **compound rest** and friction of workpieces.

You can do operations like:

- **Facing** (flat cut at the end)

- Set **the Tool Post on the square Turret,** set the angle for facing.
- To make the face smooth and reduce the length of workpieces.

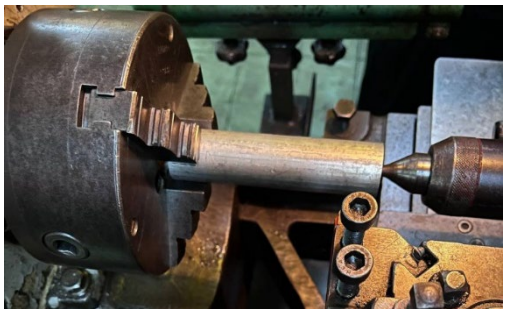


- **Turning** (reducing diameter)

- Use the tailstock to support the workpieces when we want to do turning.
- For Carbon Steel, cutting it slowly with a low depth
- For Aluminum, cutting it with a higher depth depending on the cutting tool.

- **Drilling** (with drill bit in tailstock)

- Change the type of the tailstock to the drill bit for drilling.
- Then move it slowly after it touches the workpieces.



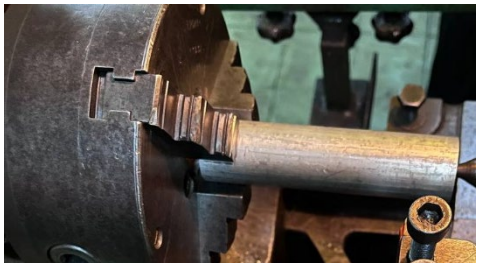
6. Stop the Machine

- Move the cutting tool out and far from the workpieces
- Touch **the button off** when you need to stop
- Push **the brake** down when we want to stop it in any sort of time or emergency situation

7. Remove Workpieces

- Remove the workpiece after it fully stops.
- Move **the Tail Stock assembly** out
- Take the specimens out of **the Chuck**
- Use water or oil to clean the specimens

The result that we get from the test:



Tips:

- Use **cutting oil** or **water** for a better finish and tool life, to reduce heat and friction, to make it clean, and to help our specimens be smooth.
- Always **disengage the half-nut** only when the tool is retracted.
- Measure the dimensions of the specimens.

- Look out for the tool cutting.
- Wear protective equipment to protect yourself from damage to the machine.
- Watch out when you start the machine.

IV. Conclusion

This lathe machine practice helped us understand how to set up and use the machine safely and accurately. We learned how to:

- Know and understand each part of the machine
- Know the material and the cutting that you need to choose
- Can use the machine to do facing, drilling, and turning.
- Know the process of a machine like

  - Set up the workpieces
  - Install the cutting tool
  - Select the speed & feed
  - Engage the motion
  - Perform Machining Operation
  - Stop the machine
  - Remove the workpiece

So, a lathe machine is a versatile machine tool that removes material from a rotating workpiece to shape it into desired forms. We can use this machine to reduce the dimensions, length, and drill the holes of workpieces. To gain the skill, the student should spend a lot of time practicing and working with the machine.

V. References

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